

1. What is the primary goal of data anonymization?

To remove or alter identifying information so an individual cannot be identified from the dataset, while still keeping the data useful for analysis.

2. What is the Havasupai Tribe case about?

A research ethics case where blood samples given for diabetes research were later used for other studies without proper consent, violating trust and informed consent.

3. What was the largest health data breach in history?

The Anthem (2015) breach, which exposed personal/health-related information for a very large number of people.

4. What is the pulse oximeter controversy about?

Pulse oximeters can be less accurate for darker skin tones, meaning they may miss dangerously low oxygen levels more often in Black patients due to bias in device design/testing data.

5. What is the role of an Institutional Review Board (IRB) in research?

To review and monitor research involving human subjects to protect participants, checking risks vs benefits, ensuring informed consent, and requiring confidentiality safeguards.

6. What is the difference between spreadsheet software and database software for health data management?

Spreadsheets: best for smaller/simple datasets; easy manual entry; limited structure, access control, and error prevention.

Databases: built for larger/complex datasets; support multiple related tables, powerful queries, better integrity, and controlled access.

7. What is a conflict of interest in public health research?

A situation where secondary interests (money, career, personal ties, institutional pressure) could bias or appear to bias research design, analysis, or reporting.

8. What is informed consent in the context of health data collection?

When a person voluntarily agrees to data collection/participation after being clearly informed about purpose, procedures, risks/benefits, how the data will be used/shared, and their rights (including withdrawal when applicable).

9. What is data masking in the context of de-identification of health data?

Hiding or replacing sensitive values (like names, IDs) with altered values (ex. fake IDs) so the original identity isn't directly visible.

10. What is the principle of k-anonymity in the context of de-identification of health data?

Each record should be indistinguishable from at least $k-1$ other records based on "quasi-identifiers" (like age/ZIP), reducing the chance of re-identification.

11. What is the main concern about conflicts of interest in public health research?

They can distort scientific judgment, leading to biased results, harmful decisions/policies, and loss of public trust.

12. What is the first step in managing conflicts of interest in public health research?

Disclosure/transparency, identify and declare the conflict early (to the IRB, institution, and/or in publications).

13. What is the role of independent oversight in managing conflicts of interest in public health research?

An independent body (IRB, committee, external reviewers) helps ensure decisions aren't driven by conflicted parties by reviewing methods, monitoring conduct, and enforcing safeguards.

14. What is one strategy for mitigating conflicts of interest in public health research?

Create a management plan (e.g., limit the conflicted person's role in data analysis/decisions), require disclosure, or in stronger cases divest/recuse.

15. What is the potential impact of not properly managing conflicts of interest in public health research?

Biased or unreliable findings, bad health decisions, reputational damage, legal/ethical violations, and reduced willingness of the public to share data or participate in research.